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https://doi.org/10.1016/j.tsf.2025.140784

PUBLISHER

Elsevier B.V

VERSION

VoR (Version of Record)

PUBLISHER STATEMENT

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REPOSITORY RECORD

Infante-Ortega, Luis, Xiaolei Liu, Kieran Curson, Luksa Kujovic, Mustafa Togay, Adam Law, Ali Abbas, Patrick Isherwood, Jake Bowers, and Michael Walls. 2025. “Thin Film Indium Tin Oxide Transparent Conductors: Comparing the Effects of Deposition Temperature and Annealing”. Loughborough University. https://hdl.handle.net/2134/30501188.v1.

Thin Solid Films 827 (2025) 140784



Contents lists available at ScienceDirect

Thin Solid Films

journal homepage: www.elsevier.com/locate/tsf



Thin film Indium Tin oxide transparent conductors: Comparing the effects of deposition temperature and annealing

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ARTICLE INFO

Keywords:
Sputtering
Thin film
Transparent conducting oxide
Indium tin oxide
Heat treatment

ABSTRACT

Indium Tin-Oxide (ITO) transparent conducting films were prepared by RF magnetron sputtering on glass substrates. The effects of two different heat treatments – annealing and heating the substrate, both at 300 °C and 500 °C – were examined to determine how they affect the microstructure, optical and electrical properties of the films. Despite extensive work on both heat treatment methods in literature, the two have not been compared in a side-by-side study thus far, and this work aims to address this. Heating the substrate during deposition produced lower resistivity films with higher carrier concentrations and higher transmission over visible wavelengths. Microstructural analysis revealed that both heat treatments produced crystalline ITO films, but the preferred crystal orientations were different. Annealing resulted in a preferred cubic (222) orientation, and high substrate temperatures yielded a mixed phase crystal structure (cubic and rhombohedral) with a preferred cubic (211) orientation. Scanning Transmission Electron Microscopy (STEM) analysis showed that sputtering at room temperature with no added heat produced films with a mostly amorphous structure which were converted into equiaxed grains by annealing. High substrate temperatures resulted in a columnar grain structure. These results have useful implications for the deposition of ITO films for various opto-electronic applications.

1. Introduction

Indium Tin-Oxide (ITO) is a degenerate n-type semiconductor with a bandgap above 3.7 eV and is a widely used transparent conducting oxide (TCO) due to its low electrical resistivity and high transmission of visible light wavelengths. These properties make it an important contact in numerous opto-electronic applications such as liquid crystal displays (LCDs) [1], thin film photovoltaics [2,3] and electrochromic devices [4].

ITO can be deposited by numerous methods including pulsed laser deposition (PLD) [5], dip coating [6], ion beam sputtering [7] or reactive thermal evaporation [8] but it is commonly prepared by magnetron sputtering using either radio frequency (RF) or direct current (DC) power [9]. This process is widely used in industry because of its reproducibility, uniformity, fast deposition rates, high quality films, ability to sputter at low pressures, excellent film adherence and the ability to sputter onto large areas. Magnetron sputtering of ITO films is important for many optoelectronic devices and has become a well-established mass production industrial process [10,11].

ITO tends to be amorphous when sputter-deposited at low substrate temperatures (<200 °C). As a result, films contain defects and are non-stoichiometric in composition, which hinders their optical and electrical properties [12]. Thin films of ITO are commonly heat treated using post-deposition annealing, which exposes the thin film to high temperatures either in air or in a vacuum. This promotes grain growth and crystallinity in the film structure, reducing defects and improving opto-electrical properties. Annealing oxidises non-stoichiometric compositions and rearranges the atoms, resulting in polycrystalline films with reduced surface roughness, decreased resistivity, and increased transmission of visible wavelengths of light [12,13]. Annealing also improves the carrier concentration and mobility as the temperature is increased, resulting in lower sheet resistance [14].

Another common heat treatment for ITO films consists of heating the substrate during the deposition of the thin film. This results in a polycrystalline microstructure with improved optical and electrical

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https://doi.org/10.1016/j.tsf.2025.140784
Received 4 October 2024; Received in revised form 28 August 2025; Accepted 29 August 2025
Available online 30 August 2025
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properties [15]. Substrate temperatures above 200 °C increase the grain size and lower sheet resistance. Increasing substrate temperatures will cause lattice expansion in the ITO structure which increases carrier concentration, as well as augmenting grain size, leading to higher carrier mobility. This heightened electron concentration causes the films to have lower transmission for infrared (IR) wavelengths [16]. The substrate must be heated above a certain temperature threshold, so the film grows layer-by-layer rather than by three-dimensional (3D) nucleation [17,18]. The crystallinity and orientation of the resulting film depend on the substrate temperature used [9,19].

The effects of both heat treatments on ITO are well understood individually, but despite extensive work on both individual processes, literature lacks a direct side-by-side comparison of how the two processes affect the optical, electrical, and microstructural properties of ITO thin film. This study aims to address this deficiency and will show how the film microstructure evolves as the processing temperature is increased and determine which heat treatment produces films with the highest optical transmission across the visible light spectrum.

2. Experimental procedure

Five ITO thin films were sputter-deposited using an AJA ATC 2200 magnetron sputtering system using 200 W of RF power on a 4-inch diameter, 0.125-inch-thick ceramic ITO compound target with a composition of 90 wt% indium oxide (In_2O_3) and 10 wt% tin dioxide (SnO_2) which was supplied by Plasmaterials Inc, USA. These conditions provided a relatively low power density of 2.47 W.cm^{-2}. This was used since high power densities may result in less transparent films with higher refractive indices and extinction coefficients [20]. All films were prepared with 0.56 % oxygen in the sputtering atmosphere, as initial testing (outlined in Section 3.1) showed this achieved ITO films with low resistivity and high transmission.

All thin films were deposited onto 5 cm x 5 cm x 1 mm thick soda-lime glass (SLG) substrates which had been cleaned in an ultrasonic bath using detergent (Decon90, Decon Laboratories, active ingredient Potassium Hydroxide (KOH)) mixed with DI water in a 5 % solution, followed by acetone, and Isopropanol alcohol (IPA). Samples were rinsed with DI water between stages, and all cleaning steps were carried out without additional heating. SLG was chosen for this investigation as it has a high softening point (>700 °C) [21] which is well suited for the temperature ranges tested. During deposition the substrates are housed in a sample holder placed 18.4 cm above the target which rotates at 10 rotations per minute (rpm) around its own axis. These conditions are used in every deposition to ensure film uniformity. Copies of films previously made were often deposited to ensure homogeneity and to check that maintenance such as target changing or cleaning had not altered the resistivity of the films. Working gas pressure was kept at 1 mTorr (0.133 Pa) during the coating process.

The ITO thin films were prepared using different heat treatments as shown in Table 1. Sample 1 was a control film prepared without any heat treatments during or after deposition, here referred to as 'room temperature' (RT). Samples 2 and 3 were prepared identically to sample 1 but were annealed post-deposition under vacuum for one hour at 300 °C or 500 °C respectively. No gases were used during the anneal. Vacuum annealing was performed in the same chamber as the deposition at a pressure of 2 × 10^{-7} Torr (2.67 × 10^{-5} Pa). Samples 4 and 5 were deposited while the substrate was heated at substrate temperatures of 300 °C and 500 °C, respectively. Substrate temperature was monitored using a built-in thermocouple attached to the substrate holder.

Table 1
Heat treatments applied to the ITO films during or after deposition.

Sample No.	Sputtering Temp (°C).	Annealing Temp (°C).
1	RT	Not annealed
2	RT	300
3	RT	500
4	300	Not annealed
5	500	Not annealed

The deposition rate for each set of conditions was determined using an Ambios XP-2 profilometer to measure thin film thickness and deposition rate. The deposition rate was used to calculate the time needed to make all films 500 nm thick. Under these conditions the films prepared at RT had a deposition rate of 381 nm/hr. Heating the substrate accelerated the deposition rate, with samples 4 and 5 having deposition rates of 430 nm/hr and 425 nm/hr respectively.

The optical properties of the films were measured using a Varian Cary 5000 UV–VIS-NIR spectrophotometer with an integrating sphere attachment, across wavelengths ranging from 200 nm to 2500 nm, with a step size of 1 nm. These measurements gave the average transmission and reflection values across visible light wavelengths (400 to 700 nm), shown in Table 2. The direct bandgap of each film was obtained via extrapolation from Tauc plots created using the transmittance data.

The electrical properties of the films are summarised in Table 3. Sheet resistance was measured using a Jandel four-point probe, with five measurements taken across the film's surface and averaged to obtain the final value. Subsequently, 1 × 1 cm pieces were cut from the centre of each sample, and metal alloy contacts were soldered onto each corner using an ultrasonic soldering iron. Hall effect measurements were then performed in the dark using the Ecopia HMS-3000 system with the Van der Pauw configuration to extract conductivity, carrier concentration, and mobility. A constant magnetic field of approximately 0.55 T is applied perpendicular to the sample while a current is passed through it. The resulting Hall voltage, generated due to the Lorentz force acting on charge carriers, indicates the carrier type based on its polarity. Carrier concentration is then determined from the Hall voltage, applied current, magnetic field strength, and sample thickness.

X-ray diffraction (XRD) peaks were obtained using a Bruker AXS D2 Phaser system with copper Kα radiation from the X-ray source. Measurements were taken over a 2θ range of 10–80° Measurements lasted 50 min, after which background noise was removed and peaks were matched to an International Centre for Diffraction Data (ICDD) Powder Diffraction File (PDF) database to identify phases and grain orientation. Cross-sectional Scanning Transmission Electron Microscopy (STEM) analysis was performed using an FEI Tecnai F20 system. The samples were prepared by Focused Ion Beam (FIB) milling using a dual beam FEI Nova 600 Nanolab. An in-situ lift out method was used to prepare the cross-sectional samples. The Tecnai F20 system is fitted with both a High Angular Annular Dark Field (HAADF) and a Bright Field (BF) detector, and images were taken using an operating voltage of 200 kV.

3. Results

3.1. Optimisation of deposition conditions

The properties of ITO films are highly dependent on the oxygen content in the sputtering atmosphere [22]. Oxygen is introduced into the argon working gas through a mass flow controller during the deposition

Table 2
Heat treatments applied to the ITO films during or after deposition.

Sample No.	Heat treatment	Average transmission (%)	Average reflection (%)	Bandgap (eV)
1	None	75.11	17.83	3.83
2	Annealed 300 °C	81.58	16.08	4.00
3	Annealed 500 °C	82.77	15.35	4.09
4	Sputtered 300 °C	83.28	13.46	4.22
5	Sputtered 500 °C	82.85	13.56	4.26

Table 3
Effects of annealing heat treatments at 300 °C and 500 °C on the electrical properties of ITO thin films.

Sample	Heat treatmen	Sheet resistance (Ω/sq)	Carrier conc. (cm^{-3})	Mobility (cm^2vs^{-1})	Conductivity (Ωcm^{-1})	P/N type:
1	None	11.2	3.79×10^{20}	32.9	2.00×10^3	N
2	Annealed 300 °C	17.0	4.26×10^{20}	19.2	1.31×10^3	N
3	Annealed 500 °C	7.0	7.34×10^{20}	29.1	3.41×10^3	N
4	Sputtered 300 °C	3.7	8.95×10^{20}	41.8	5.78×10^3	N
5	Sputtered 500 °C	2.9	1.15×10^{21}	40.5	7.46×10^3	N

to improve the optical and electrical properties of the films. Depositing ITO thin films in a pure argon atmosphere can result in significant transparency losses [23] and using too little oxygen can lead to delamination [24]. Conductivity is reduced when the oxygen content is too low, resulting in poor film structure, or too high, as oxygen vacancies are filled and carrier concentration is reduced [25]. The oxygen flow must be carefully controlled to optimise the optical and electrical properties.

A series of 500 nm thick ITO thin films were deposited using a range of oxygen flow rates to optimise the properties of the thin films. The concentration of oxygen in the sputtering atmosphere ranged from 0.45 % to 0.79 %. Fig. 1 shows the resulting optical and electrical properties. Using a 0.56 % oxygen concentration in argon resulted in a film with the highest optical transmission in the visible wavelengths while maintaining a low sheet resistance. This oxygen concentration was then used to prepare thin films 1–5, shown in Table 1, and was achieved using 3 SCCM of gas flow from a 5 % oxygen-in-argon supply, and 24 SCCM from a pure argon line. The sharp decrease in transmission when using 0.79 % oxygen may be explained by the findings of Bender et al [26], who found that increasing the oxygen flow when sputtering ITO will lower the bandgap of the film. Gas flows into the sputtering chamber were controlled using mass flow controllers.

3.2. ITO deposited at room temperature

Fig. 2 shows the transmission and reflection of sample 1, which was deposited at room temperature and not subjected to any annealing heat treatments. Sputtering at RT resulted in an average transmission of 75.1 % and an average reflection of 17.8 %, respectively. The bandgap was calculated to be 3.83 eV.

The sheet resistance across the film surface averaged 11.2 Ω/sq. Hall Effect measurements showed a carrier concentration of 3.79×10^{20} cm^{-3} and carrier mobility of 32.9 cm^2vs^{-1}.

XRD analysis (Fig. 3) shows that the film sputter-deposited at RT had

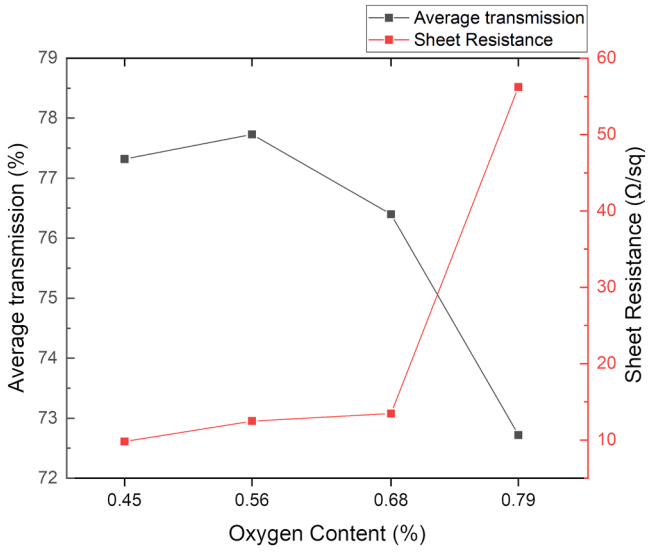


Fig. 1. Average transmission (400 – 700 nm) and sheet resistance of ITO films sputtered at different oxygen concentrations in the working gas.

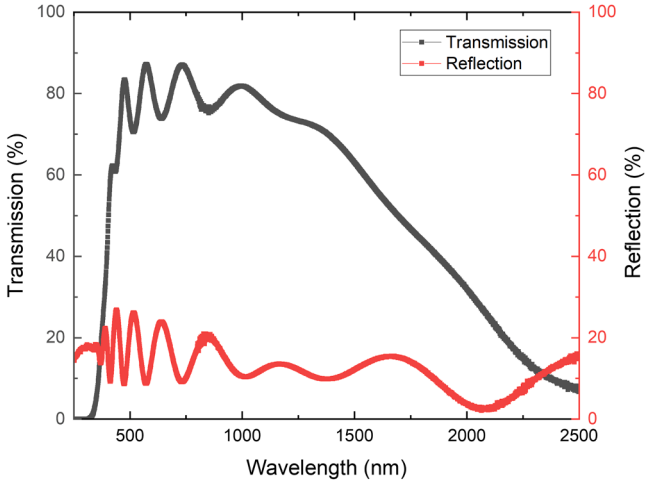


Fig. 2. Transmission and reflection spectra of 500 nm ITO film deposited at room temperature with no annealing.

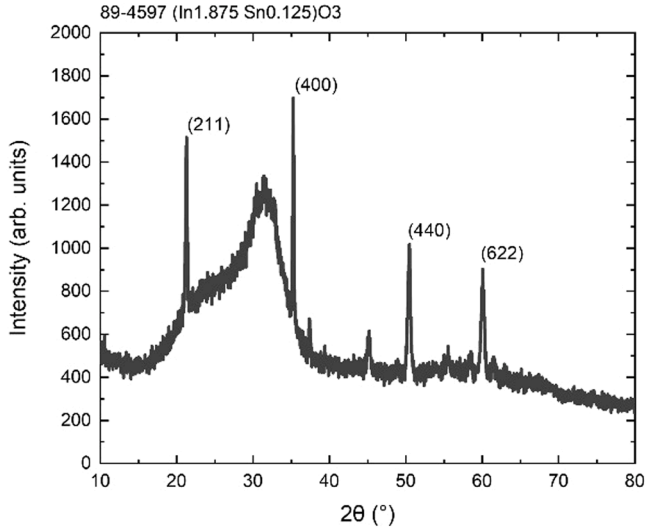


Fig. 3. XRD pattern of ITO film sputtered at room temperature.

a mostly amorphous XRD profile, and a few pronounced peaks were detected which indicate partial crystallinity. The cross-sectional STEM analysis (Fig. 4) supports this, showing a predominantly amorphous structure with few nanocrystalline areas.

3.3. Annealed ITO thin films

Two samples of the ITO film deposited at room temperature and described in Section 3.2 were annealed for 1 h at 300 °C and 500 °C respectively. Both annealing temperatures improved the optical properties significantly (Fig. 5). The higher temperature anneal resulted in the largest improvement, increasing the transmission by 7.7 % and



Fig. 4. Cross-sectional STEM analysis of ITO film sputter-deposited at room temperature, showing a mostly amorphous microstructure.

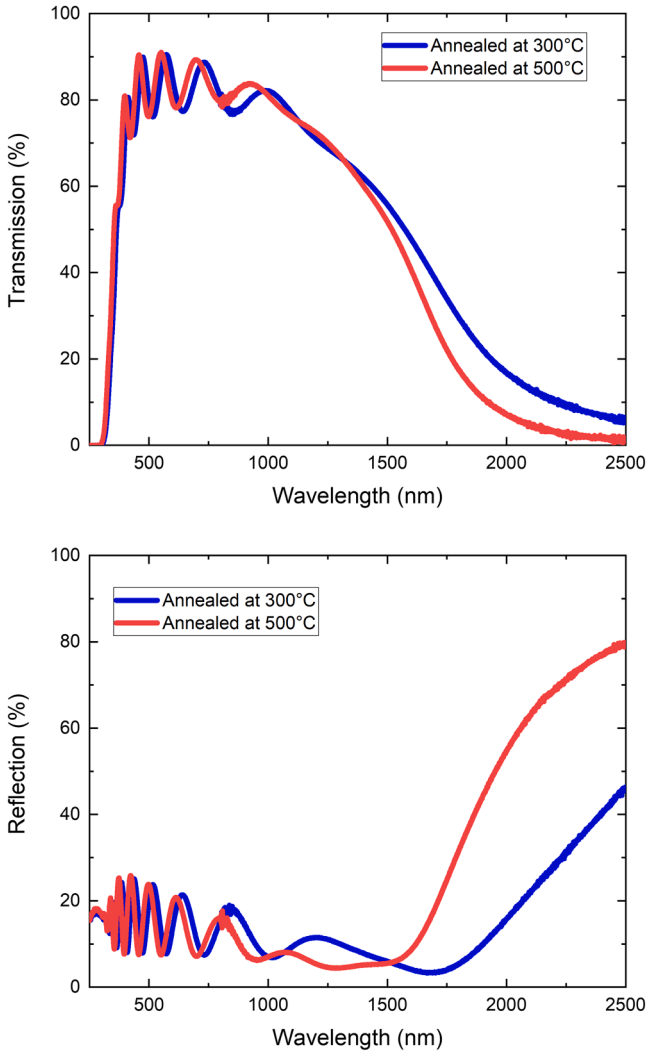


Fig. 5. Transmission (top) and reflection (bottom) spectra of ITO films annealed at 300 °C and 500 °C.

reducing the reflection by 2.5 %.

The 500 °C anneal significantly improved film conductivity. The sheet resistance reduced to 7 Ω/sq and the carrier concentration was almost doubled. However, the 300 °C anneal actually increased the sheet resistance by 6 Ω/sq and reduced both the conductivity and carrier mobility. This was repeated several times with fresh depositions and always produced the same result. Hu et al [12] observed a similar result when annealing ITO films under vacuum, finding that carrier mobility can act erratically near the amorphous-crystalline transition point. Both annealing temperatures also resulted in an increase in the bandgap to over 4 eV.

Both the annealing treatment temperatures of 300 °C and 500 °C resulted in the formation of pronounced crystalline peaks in the XRD spectra pattern, as shown in Fig. 6. The annealed films have a cubic structure peak matched to PDF 89–4597. The strongest lattice peak for the annealed films showed a preferential orientation in the (222) direction. STEM cross-sectional analysis showed the films had a dense grain structure with no evidence of voiding. Both annealed films showed an equiaxed grain structure, but the grain size in the 500 °C anneal was much larger than for the 300 °C sample Fig. 7.

3.4. ITO deposited at high substrate temperatures

The transmission and reflection spectra of ITO films prepared using substrate temperatures of 300 °C and 500 °C are shown in Fig. 8. Both these films had more desirable optical properties than the film prepared at RT, with the highest average transmission of all films (83.3 %) achieved using a substrate temperature of 300 °C. Heating the substrate also produced films with the widest bandgaps: 4.22 eV and 4.26 eV for 300 °C and 500 °C, respectively.

High substrate temperatures also produced the films with the lowest sheet resistance and highest carrier concentrations. Heating the substrate to 500 °C during deposition produced ITO films with an average sheet resistance of 2.9 Ω/sq and carrier concentration of 1.15×10^{21} cm^{-3}.

The XRD pattern in Fig. 9 shows that substrate heating produces films with pronounced crystalline peaks. The XRD pattens show a mixture of cubic (PDF 89–4597, peaks labelled in black) and rhombohedral (PDF 89–4599, peaks labelled in orange) crystal structures, with a dual peak around $2\theta = 30°$ Vink et al [27] found that this same dual crystal structure can appear when sputter-depositing ITO and depends on the oxygen partial pressure and heat treatment used during deposition. Films made at high substrate temperatures had a preferred (211) grain orientation around $2\theta = 22°$ regardless of temperature. The rhombohedral grain structure of the films deposited at high substrate temperature showed a preferred orientation in the (003) direction.

High substrate temperatures also slightly shifted the XRD peaks to the left. Peak shifting can occur if stress is present in the films [28]. Stress can lead to delamination in extreme cases [24]. STEM cross-sectional analysis (Fig. 10) showed that high substrate temperatures resulted in a distinct columnar grain structure with large grains and no evidence of voiding.

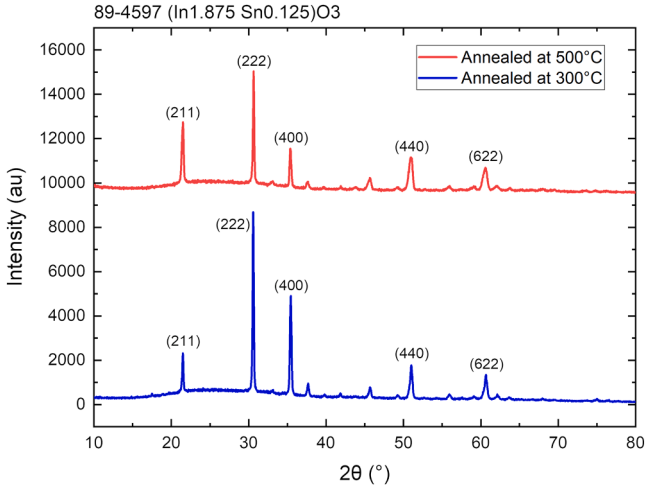


Fig. 6. XRD patterns of ITO films annealed at 300 °C and 500 °C.

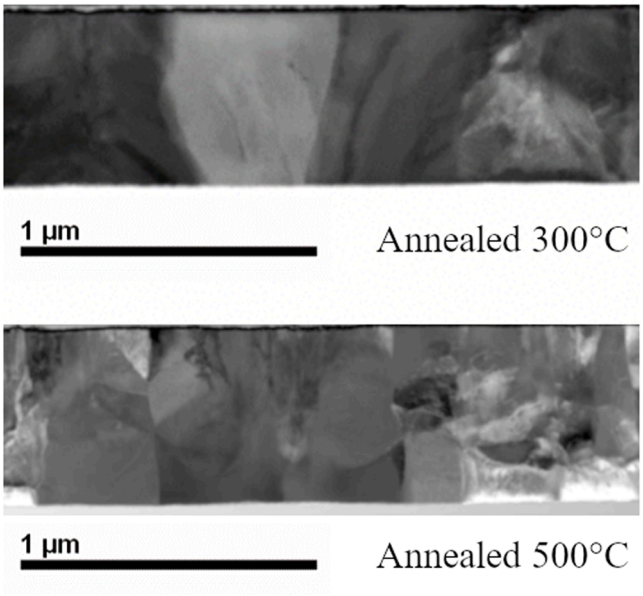


Fig. 7. Cross-sectional STEM analysis of ITO films annealed at 300 °C (top) and 500 °C (bottom). Annealing causes the microstructure to change from amorphous to crystalline.

4. Analysis

Exposing the thin film ITO to substrate heating or annealing results in crystalline films in both cases, but the two processes produce different material properties and microstructures.

Compared to the as-deposited RT film, all heat treatments resulted in significantly increased transmission at lower wavelengths. The ITO thin films deposited at high substrate temperatures outperformed their annealed counterparts in terms of optical transmission in the visible range from 400 to 700 nm. Table 2 shows that heat treating the films at 300 °C resulted in 1.7 % higher transmission and 2.62 % lower reflection in visible wavelengths when heating the substrate compared with annealing. Both heat treatments at 500 °C resulted in near identical transmission, but the average reflection was 1.79 % lower when using substrate heating. The films deposited with substrate heating also transmit low wavelength photons (~370 nm) but films which were annealed had higher transmission for wavelengths over 1000 nm.

Heat treating ITO thin films can decrease transmission in the near infrared (NIR) wavelengths of light. Transmission in this region is determined by the absorption of free carriers, which increases when the films are heat treated. This phenomenon is often explained by the Drude theory of free charge carriers [12,17,29,30]. Results obtained in this study support this. A decrease in transmission at wavelengths above 1000 nm was observed in all heat-treated ITO thin films, which previous work [28] attributes to the heat treatments increasing the number of free charge carriers, i.e., the carrier concentration in the films. This increase in carrier concentration was confirmed by Hall effect measurements as shown in Table 3. The effect is more pronounced when using high substrate temperatures than when annealing.

High substrate temperatures during deposition also resulted in a very pronounced increase in reflection for wavelengths over 1250 nm. This can have useful applications. For example, spectrally selective coatings for photovoltaic devices allow the transmission of visible wavelengths of light while reflecting IR wavelengths. This is useful because the power output of photovoltaic modules is reduced when operating at high temperatures, and photons beyond the bandgap of the absorber material can cause the module temperature to increase [31]. Low-emissivity, or 'low-E' glass also uses this principle to retain heat in buildings and often incorporates an ITO coating [32], and the concept also has uses in

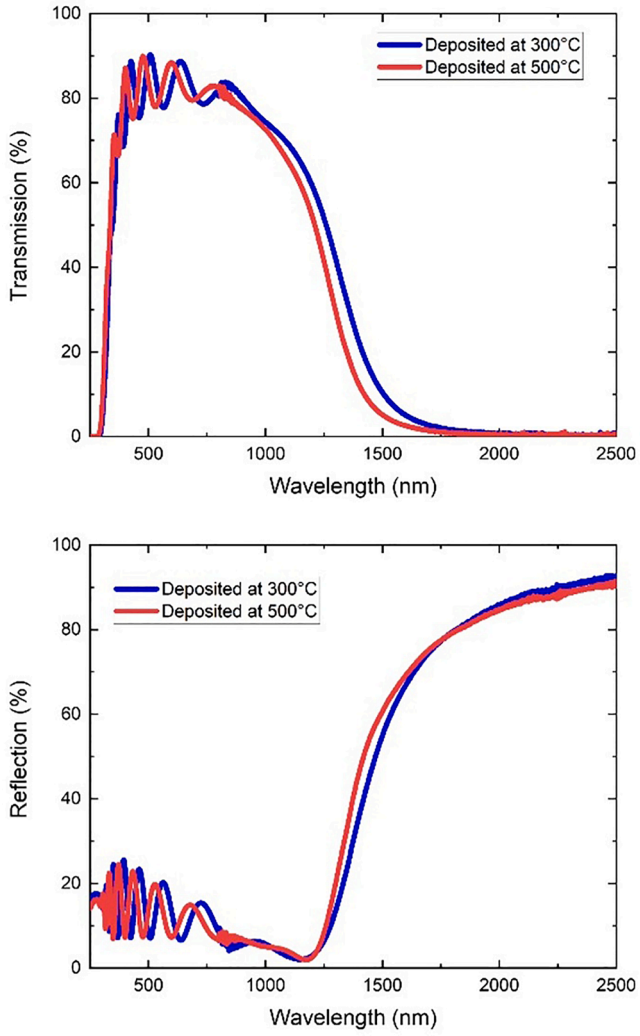


Fig. 8. Transmission (top) and reflection (bottom) spectra of ITO films sputtered with substrate temperatures of 300 °C and 500 °C.

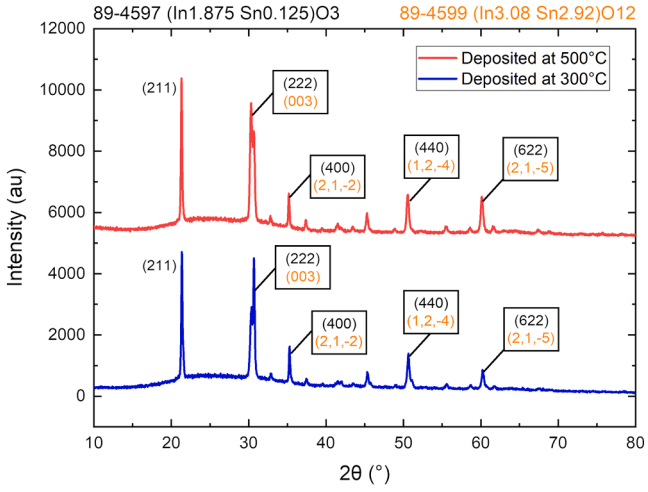


Fig. 9. XRD patterns of ITO films sputtered with substrate temperatures of 300 °C and 500 °C.

greenhouses for thermal management of crops [33].

A direct bandgap increase is observed after all heat treatments. The increase in bandgap is highest when using substrate heating. The

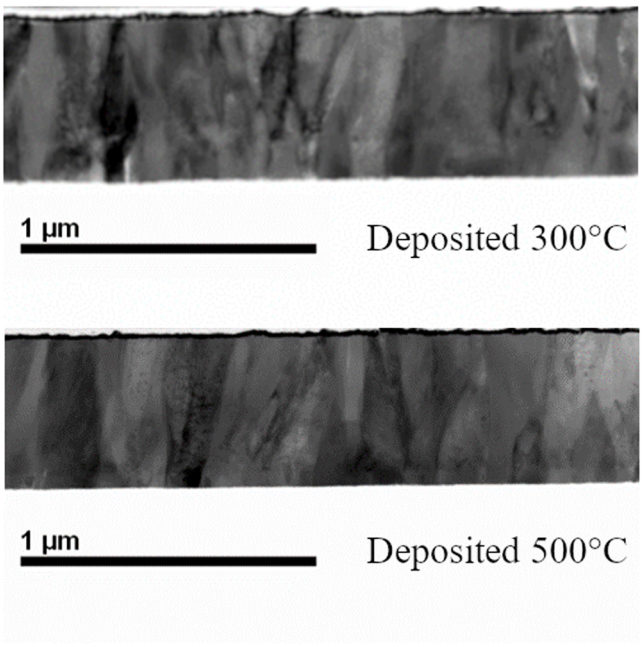


Fig. 10. Cross-sectional STEM analysis of ITO films sputtered with substrate temperatures of 300 °C (top) and 500 °C (bottom). The microstructure is columnar.

absorption edge of the ITO thin film deposited at RT was at longer wavelengths than the heat-treated films, likely due to the presence of a high concentration of crystalline defects that smear the band edges, leading to a lower bandgap. Annealing increased the optical transmission and lowered the absorption edge, enabling shorter wavelengths of light to be transmitted [24]. Using high substrate temperatures led to wider bandgaps than annealing. This resulted in the lower absorption edges shown in the transmission spectra in Fig. 11. This effect is consistent with previous observations [9]. Despite each heat treatment resulting in a very distinct bandgap increase, the Tauc plot method provides an estimation, and it is possible that the glass beneath the films (which has a bandgap of 4.2 eV) is interfering with the value.

The increase in bandgap is partly due to the Moss-Burstein effect, which is where the increased doping level from the increased carrier concentration fills states in the conduction band, pushing the Fermi level to a higher energy, thus increasing the bandgap. High substrate temperatures also increase conductivity, with grain size increasing significantly as the deposition temperature increases, reducing grain boundary scattering, and increasing carrier mobility [9].

The electrical properties of ITO films are strongly dependent on film microstructure [29], which is affected by the heat treatment and substrate temperature. The conductivity of amorphous ITO thin films is reduced by lack of long-range order in the microstructure. As the film becomes crystalline, the carrier concentration increases, [12].

All heat treatments used in this work produced crystalline films, with higher transmission in visible wavelengths than the amorphous films. Photons are scattered by either ionized impurity or grain boundary scattering in the ITO film [12]. Grain boundary scattering is the dominant mechanism, and as all heat treatments increase the grain size, light will be scattered less by decreasing the grain boundary length.

All heat treatments increased the conductivity of the films, except for the 300 °C anneal. Table 3 shows that heating the substrate resulted in better conductivity than annealing by every measure, regardless of temperature.

XRD analysis shows clear differences in the film microstructure depending on the heat treatment used. Annealing and increasing substrate temperature exert a distinct influence on grain growth. Annealed films have a cubic structure with a strong preferential orientation in the

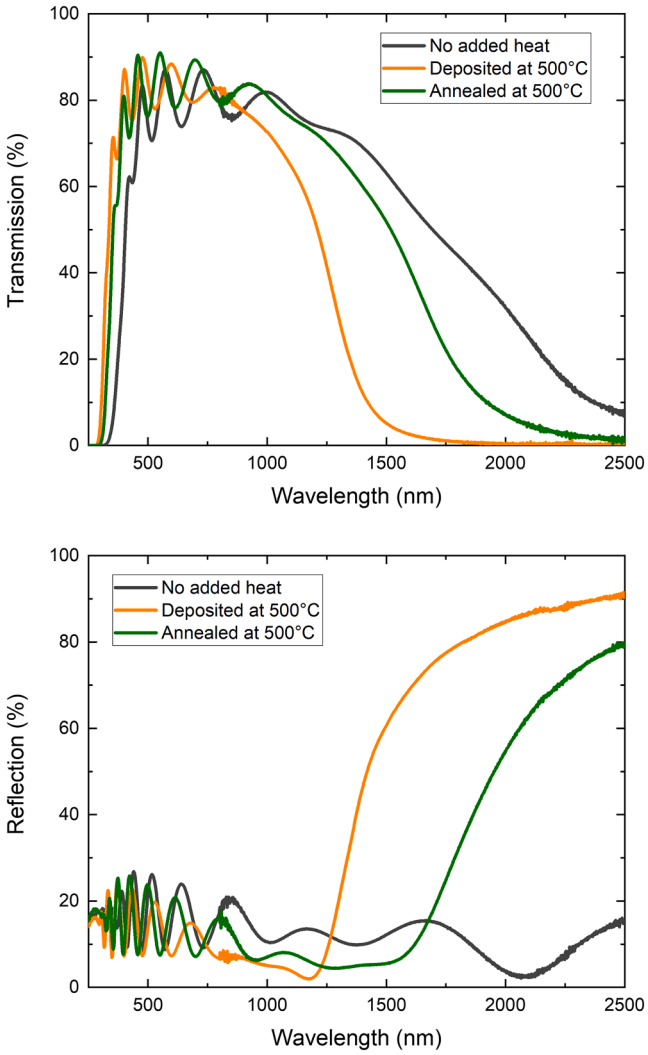


Fig. 11. Comparison of transmission (top) and reflection (bottom) spectra of ITO films prepared with substrate heating and annealing at 500 °C.

(222) direction, while films deposited with substrate heating have a mixed cubic and rhombohedral structure with (211) and (003) orientations for each phase. Carrier mobility in polycrystalline ITO films will be controlled by the scattering mechanism, which is determined by the grain structure [29]. Despite the mixed crystal structure, the grain structure achieved using high substrate temperatures results in higher carrier mobilities and carrier concentration, and overall higher conductivity than annealing.

The differences in the XRD patterns occur because the structure of the ITO film growth depends on the heat treatment used. The annealed films are predominantly amorphous or nanocrystalline and transition to crystalline during annealing. Amorphous ITO forms via a 3D nucleation in which the film grows via the coalescence of islands from nucleation sites. Films prepared while heating the substrate, are polycrystalline from the start and grow laterally in a two-dimensional (2D) manner [17, 18].

STEM cross-sections showed that annealing under these conditions results in an equiaxed grain structure. The grain size increased with temperature. The films deposited at high substrate temperatures showed clear columnar grain structures with large grains. The results highlight how the optical and electrical performance resulting from each type of heat treatment arise from the fundamentally different grain structure created.

5. Conclusion

The optical, electrical, and microstructural properties of ITO thin films deposited at high temperatures or exposed to heat treatments were examined. Depositing at high substrate temperatures resulted in increased conductivity and carrier concentration, with the highest values observed in thin films deposited at 500 °C. These thin films also exhibited higher transmission in visible wavelengths compared to annealing under vacuum, regardless of the temperature used. The highest average transmission was obtained by the film deposited at a 300 °C substrate temperature. All heat treatments resulted in an increased bandgap which was more pronounced when using high substrate temperatures, lowering the absorption edge and enabling lower wavelengths of light to be transmitted by the ITO thin film. The higher electron concentration of the substrate-heated ITO films also reduced transmission and increased reflection for NIR wavelengths. Each heat treatment yielded a fundamentally different grain structure, with columnar grains being more beneficial for optical and electrical properties. XRD analysis revealed that the films deposited at high substrate temperature had a mixed (cubic and rhombohedral) crystal structure. The XRD peaks were observed to shift due to stress in the film.

The choice of process conditions for the deposition of ITO transparent conducting layers will depend on the application. For example, the amorphous structure of the ITO deposited at room temperature will be useful when depositing on flexible substrates or polymer surfaces. However, any application that requires high transmission and conductivity will need some sort of heat treatment, and results from this study show that, under the conditions tested, high substrate temperatures yield superior results. Annealing is useful since it simplifies the deposition equipment because substrate heating and temperature control is avoided, but the final film will have poor transmission and conductivity especially at lower temperatures. Using high substrate temperatures yielded comparable results at both 300 and 500 °C, meaning that excellent film properties do not need extremely high temperatures to manifest when using this method. Using both heat treatments in tandem (i.e. depositing ITO at high substrate temperatures and then annealing) may result in further improved film properties. This, however, lies outside of the scope of this study, as the intent here was to show which heat treatment results in more desirable properties when exposing the substrate to heat for a comparable amount of time.

CRediT authorship contribution statement

Luis C. Infante-Ortega: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Xiaolei Liu:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Kieran M. Curson:** Formal analysis, Data curation. **Luksa Kujovic:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Data curation. **Mustafa Togay:** Writing – review & editing, Formal analysis. **Adam M. Law:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Ali Abbas:** Writing – review & editing, Investigation, Formal analysis, Data curation. **Patrick J.M. Isherwood:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Jake W. Bowers:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition. **John M. Walls:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors acknowledge the use of facilities within the Loughborough Materials Characterisation Centre (LMCC). The authors are grateful to UKRI and EPSRC for funding the project through EP/S010300/1.

Data availability

Data will be made available on request.

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